

STUDENT RESULT MANAGEMENT SYSTEM

Project Report

Submitted by:

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Submitted to:

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Programming Language: Python

Project Type: Object-Oriented Programming / Console Application

Academic Year: 2026–2027

1. Introduction

The Student Result Management System is a Python console application for managing basic student academic records. It stores student name, roll number, and marks and provides menu-driven operations for adding students, updating marks, and displaying records.

The project demonstrates Object-Oriented Programming concepts including classes, objects, constructors, instance variables, class variables, class methods, lists, loops, conditions, and user input.

2. Problem Statement

Managing student marks manually can be time-consuming and difficult to update or view efficiently. This project provides a simple digital solution for adding student records, finding students by roll number, updating marks, and displaying records.

3. Functional Requirements

- Add a student with name, roll number, and marks.
- Search for a student using roll number.
- Update marks of an existing student.
- Display all stored student records.
- Show an appropriate message when a student is not found.
- Exit the application.

4. Non-functional Requirements

Usability: Simple menu-driven interface.

Performance: Fast operations for the small in-memory dataset.

Maintainability: Clear class methods and structure.

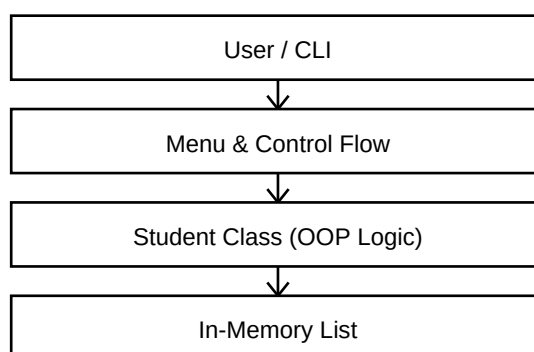
Portability: Runs on systems with compatible Python.

Reliability: Clear success and error messages.

5. System Architecture

The user interacts with the command-line menu. Menu choices invoke methods of the Student class. Student objects are stored in the class-level all_students list. No external database is used.

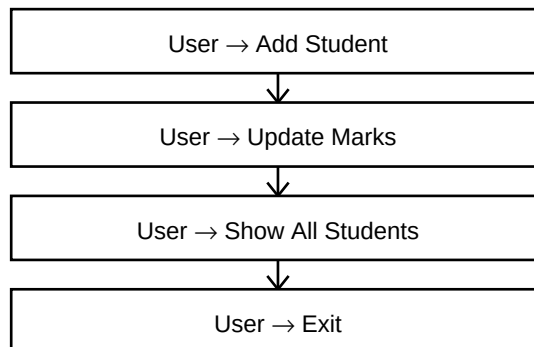
System Architecture



6. Design Diagrams

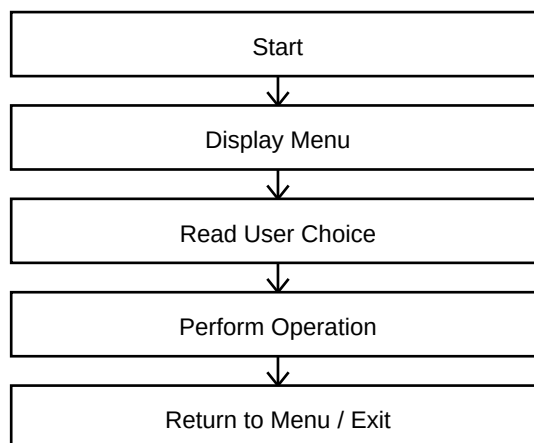
6.1 Use Case Diagram

Use Case



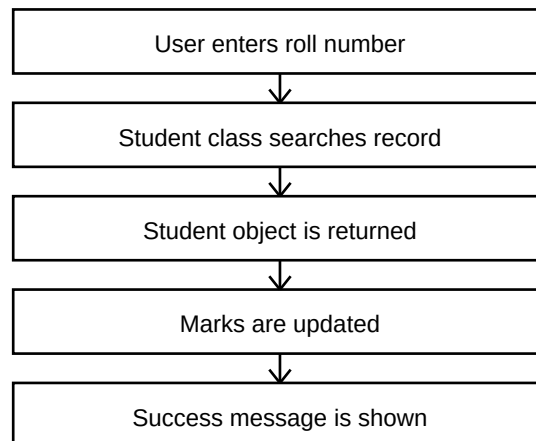
6.2 Workflow Diagram

Workflow



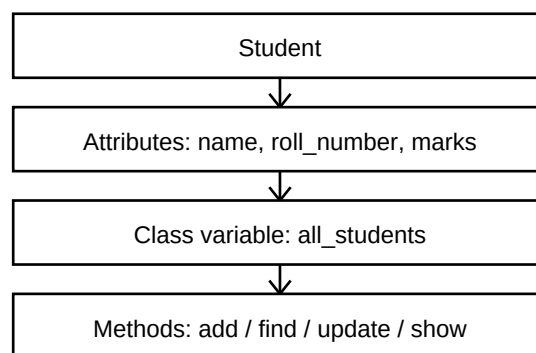
6.3 Sequence Diagram

Sequence – Update Marks



6.4 Class / Component Diagram

Student Class



6.5 ER Diagram

Not applicable in the current version because no database or persistent storage is used. Records are stored temporarily in a Python list.

7. Design Decisions & Rationale

OOP was selected to group student data and related operations. A class variable stores the shared collection of Student objects. Class methods are used for operations on that collection. In-memory storage keeps the project simple for an academic demonstration. A menu-driven interface makes testing and demonstration straightforward.

8. Implementation Details

The Student class uses `__init__` to initialize student data. `add_student()` creates and stores a Student object. `find_student_by_roll()` searches the list. `update_student_marks()` changes marks after a successful

search. `show_all_students()` displays records. A while loop repeatedly displays the menu until Exit is selected.

Python concepts used: classes, objects, `__init__`, `self`, class variables, `@classmethod`, `cls`, lists, `append()`, `input()`, `int()`, loops, conditions, `return`, `break`, and f-strings.

9. Screenshots / Results

Add screenshots of: Main Menu, Add Student, Update Marks, Show All Students, and Student Not Found case.

10. Testing Approach

Functional testing is used. Test cases include adding a valid student, displaying records, updating marks with a valid roll number, updating an unknown roll number, showing an empty list, and exiting the program.

11. Challenges Faced

The main challenges were understanding class methods and `cls`, searching for objects before updating marks, maintaining correct Python indentation, handling missing records, and testing multiple menu operations with temporary in-memory data.

12. Conclusion

The Student Result Management System demonstrates how Python and OOP can be used to build a practical student-record application. The project can later be extended with file or database storage, input validation, authentication, result calculation, and a graphical or web interface.